

LUCIDMSK CME PROGRAM

MODULE 11 — POST-TEST

Muscle Injuries & BAMIC: Sports Imaging of Muscle Pathology

Sports MSK · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)[™]

Read each question carefully and select the single best answer. A score of $\geq 70\%$ (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1

A hamstring MRI shows a partial tear of the proximal conjoint tendon involving <50% of the cross-sectional area with intramuscular extension. BAMIC grade:

- A Grade 0
- B Grade 1
- C Grade 2
- D Grade 3

2

Which MRI sequence is most sensitive for early muscle injury detection?

- A T1 fat-sat
- B STIR (short tau inversion recovery)
- C Gradient echo (GRE)
- D Post-contrast T1

3

Proximal hamstring avulsion: complete conjoint tendon retraction >2 cm from the ischial tuberosity. Best management:

- A Conservative; physical therapy only
- B Platelet-rich plasma injection
- C Surgical reattachment to the ischial tuberosity
- D Observation; return to sport in 6 weeks

4

Morel-Lavallée lesion on MRI. This is best described as:

- A Intramuscular hematoma
- B Post-traumatic closed degloving injury — seroma/hematoma between fascia and overlying subcutaneous fat
- C Fascial tear with herniation
- D Deep vein thrombosis

5 Grade 3b muscle injury (Berlin Classification) indicates:

- A Functional disorder without structural damage
- B Minor structural muscle lesion (myofascial)
- C Moderate structural lesion; partial muscle tear
- D Severe structural lesion; subtotal or complete tear with fascia disruption

6 Muscle herniation through fascial defect is MOST commonly seen in which muscle compartment?

- A Posterior compartment of leg (gastrocnemius)
- B Anterior compartment of leg (tibialis anterior)
- C Posterior thigh (biceps femoris)
- D Posterior arm (triceps)

7 MRI finding most specific for complete proximal hamstring avulsion (vs. severe strain):

- A Extensive STIR signal in the muscle belly
- B Fluid surrounding the tendon without gap
- C Gap between the avulsed tendon and ischial tuberosity $\pm$ marrow edema at tuberosity
- D Peritendinous edema only

8 Calcific myonecrosis on MRI/CT appears as:

- A Soft tissue mass with avid solid enhancement
- B Calcified fluid-filled intramuscular lesion; rim calcification; no enhancement of lumen
- C Aggressive periosteal reaction adjacent to muscle
- D Multiple T2-bright cysts with septation

9 Snapping hip — "coxa saltans" on ultrasound. The most common cause of INTERNAL snapping hip:

- | | |
|----------|---|
| A | Iliotibial band over the greater trochanter |
| B | Iliopsoas tendon snapping over the iliopectineal eminence or femoral head |
| C | Loose body in the hip joint |
| D | Piriformis tendon |

10 BAMIC Grade 0 (type P) muscle injury on MRI:

- | | |
|----------|---|
| A | Complete musculotendinous rupture |
| B | Structural intramuscular tear; no tendon involvement |
| C | Functional disorder; no MRI structural abnormality; normal scan or only minor edema |
| D | Peritendinous edema only at the osteotendinous junction |

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	C	B	C	B	D	B	C	B	B	C

Q1

A hamstring MRI shows a partial tear of the proximal conjoint tendon involving <50% of the cross-sectional area with intramuscular extension. BAMIC grade:

Answer: C

A Grade 0

B Grade 1

✓ **Grade 2**

D Grade 3

EXPLANATION BAMIC Grade 2 = partial musculotendinous tear involving 50–99% cross-sectional area of the tendon OR intramuscular tear >2 cm. Partial tear <50% without intramuscular component = Grade 1. Complete tear = Grade 3.

Q2

Which MRI sequence is most sensitive for early muscle injury detection?

Answer: B

A T1 fat-sat

✓ **STIR (short tau inversion recovery)**

C Gradient echo (GRE)

D Post-contrast T1

EXPLANATION STIR is most sensitive for acute muscle edema and early injury — detects interstitial fluid/edema before T1 changes appear. T1 fat-sat post-contrast can add specificity for tear severity and vascular involvement.

Q3

Proximal hamstring avulsion: complete conjoint tendon retraction >2 cm from the ischial tuberosity. Best management:

Answer: C

A Conservative; physical therapy only

B Platelet-rich plasma injection

✓ **Surgical reattachment to the ischial tuberosity**

D Observation; return to sport in 6 weeks

EXPLANATION Complete proximal hamstring avulsion with >2 cm retraction (BAMIC Grade 3) is a strong indication for surgical reattachment, especially in athletic populations. Retraction makes conservative healing very unlikely.

Q4

Morel-Lavallée lesion on MRI. This is best described as:

Answer: B

A Intramuscular hematoma

✓ **Post-traumatic closed degloving injury — seroma/hematoma between fascia and overlying subcutaneous fat**

C Fascial tear with herniation

D Deep vein thrombosis

EXPLANATION Morel-Lavallée = closed degloving injury from shear forces separating subcutaneous fat from the underlying fascia, creating a fluid-filled space (hematoma, seroma, or lymphorrhea). Common over greater trochanter and proximal thigh.

Q5 Grade 3b muscle injury (Berlin Classification) indicates:

Answer: D

A Functional disorder without structural damage

B Minor structural muscle lesion (myofascial)

C Moderate structural lesion; partial muscle tear

✓ **Severe structural lesion; subtotal or complete tear with fascia disruption**

EXPLANATION Berlin Classification Grade 3b = severe structural lesion with subtotal or complete muscle tear involving the endomysium, perimysium, and fascia. Grade 1 = functional (non-structural). Grade 2 = minor/moderate structural. Grade 3a = subtotal without fascial involvement.

Q6 Muscle herniation through fascial defect is MOST commonly seen in which muscle compartment?

Answer: B

A Posterior compartment of leg (gastrocnemius)

✓ **Anterior compartment of leg (tibialis anterior)**

C Posterior thigh (biceps femoris)

D Posterior arm (triceps)

EXPLANATION Tibialis anterior herniation through the anterior crural fascia is the most commonly reported muscle herniation. Appears as a soft tissue bulge on standing/loading MRI or ultrasound. Fascial defect typically 1–2 cm diameter.

Q7 MRI finding most specific for complete proximal hamstring avulsion (vs. severe strain):

Answer: C

A Extensive STIR signal in the muscle belly

B Fluid surrounding the tendon without gap

✓ **Gap between the avulsed tendon and ischial tuberosity ± marrow edema at tuberosity**

D Peritendinous edema only

EXPLANATION Complete avulsion = gap between the avulsed tendon and ischial tuberosity on coronal/axial MRI, ± bone marrow edema at the ischial tuberosity. Retraction distance measured from the tuberosity determines surgical urgency.

Q8 Calcific myonecrosis on MRI/CT appears as:

Answer: B

A Soft tissue mass with avid solid enhancement

✓ **Calcified fluid-filled intramuscular lesion; rim calcification; no enhancement of lumen**

C Aggressive periosteal reaction adjacent to muscle

D Multiple T2-bright cysts with septation

EXPLANATION

Calcific myonecrosis = late sequela of compartment syndrome or muscle ischemia. CT/MRI shows encapsulated intramuscular lesion with peripheral rim calcification and central fluid/necrotic material with NO internal enhancement. Can simulate sarcoma — key is chronicity and prior trauma.

Q9

Snapping hip — "coxa saltans" on ultrasound. The most common cause of INTERNAL snapping hip:

Answer: B

A Iliotibial band over the greater trochanter

✓ **Iliopsoas tendon snapping over the iliopectineal eminence or femoral head**

C Loose body in the hip joint

D Piriformis tendon

EXPLANATION

Internal coxa saltans = iliopsoas tendon snapping over the iliopectineal eminence or femoral head. External snapping = IT band over the greater trochanter. Dynamic ultrasound or fluoroscopic bursal injection is diagnostic and therapeutic.

Q10

BAMIC Grade 0 (type P) muscle injury on MRI:

Answer: C

A Complete musculotendinous rupture

B Structural intramuscular tear; no tendon involvement

✓ **Functional disorder; no MRI structural abnormality; normal scan or only minor edema**

D Peritendinous edema only at the osteotendinous junction

EXPLANATION

BAMIC Grade 0 (Functional, type P = "performance") = no structural abnormality on MRI. The athlete has pain and impaired performance but MRI is normal or shows only minor physiologic edema. This is a diagnosis of exclusion — clinical evaluation critical.